Instruction Manual

K-Shear® Triaxial Accelerometers

**Types 8790A…,
8791A…,
8792A…,
8793A…,
8794A…,
8795A…**

CE

8790A_002-080e-10.05

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8795A…

CE

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1 Introduction

By choosing Kistler K-Shear triaxial accelerometers you have opted for an instrument distinguished by precision, long life and technical innovation. Please read these instructions carefully before installing and operating these instruments.

Kistler offers a wide selection of measuring instruments and comprehensive systems including:

- Quartz and ceramic sensors for force, pressure, acceleration, shock, vibration and strain.
- Associated signal conditioners and accessories.
- Piezoresistive pressure transducers and transmitters with associated amplifiers.

Kistler can also provide entire systems for special purposes in the automotive industry, plastics processing, and biomechanics. Our general catalog provides information on Kistler products and services. Detailed technical data sheets are available on most products offered.

Worldwide customer service is at your disposal should you have any questions regarding this or other Kistler products. Information on your specific application is also available from Kistler.

2 Important Guidelines

To ensure your personal safety, please observe the safety guidelines in this section.

2.1 For your safety

- Carefully follow the installation information contained in section 5 of this manual.
- Do not drop the instrument.
- Store in the case provided and in a clean, dry environment.
- Power the instrument in accordance with the instructions in section 6 of this manual.

2.2 Precautions

Kistler sensors are thoroughly tested before leaving the factory. To maintain safe and reliable operation, it is important to follow the instructions herein.

- Do not mount accelerometers on high voltage surfaces.
- Keep cable clear of power lines and open machinery.
- Never operate or store the unit beyond the specified temperature range.
- Do not exceed the maximum specified current.
- Never exceed the maximum specified voltage.
- Follow the instructions for mounting. Do not over tighten.
- Do not expose the unit to excessive shock, i.e., by using a hammer or dropping the unit.
- When not in use, store the accelerometer in the container supplied. Always store in a clean, dry area.
- Keep the connector clean and covered when not in use
- Follow the recommendations in section 7.4

2.3 Using This Manual

Information contained herein includes a technical description, installation and operating instructions, powering and considerations for cable length. Section 8 contains a listing of mounting accessories and cables available to assist with your measurement needs.

It is recommended that the entire manual be read prior to installation and operation of K-Shear triaxial accelerometers. The user who has prior experience with Piezotron® type accelerometers may want to confine reading to particular sections of interest.

We have endeavored to arrange these instructions in a manner that allows for easy location of topics of interest. Your Kistler representative is also available to assist with any questions.

Information contained herein may be subject to change.

2.4 K-Shear Accelerometer Types

This manual is applicable to all K-Shear® triaxial accelerometers listed on the title page. Please see the product data sheets packaged with this manual for detailed specifications.

2.5 Manual Nomenclature

Throughout this manual some special designations and nomenclature are used for specific terms and concepts relating to K-Shear accelerometers. These are explained in Table 1.

Term	Definition
FS	Full-Scale
TEDS	Transducer Electronic Data Sheet
PiezoSmart	The Kistler trade name for accelerometers that contain TEDS

Table 1: Manual Nomenclature

3 General Description

Kistler K-Shear triaxial accelerometers are shear mode shock and vibration measuring instruments. A self-generating piezoelectric sensing element is used in conjunction with the built-in, internal circuit Piezotron impedance converter. As with most accelerometers, the sensitivity of this series is expressed in terms of the ratio of the electrical output to applied acceleration (e.g. mV/g). In the case of Piezotron devices such as K-Shear the output is a low impedance voltage signal, which is proportional to the applied acceleration.

Being a low impedance device, no charge amplifier or special cabling is required and transmission over long lines is possible with a minimum of noise pick-up.

Accelerometers with a "T" suffix, such as 8702B…T are variants of the standard version incorporating the "Smart Sensor" design. Viewing an accelerometer's data sheet requires an Interface/Coupler such as Kistler's Model 5000M07 PDA based or 5000M04 PC (serial port) based TEDS Editor software. The Interface provides negative current excitation (reverse polarity) altering the operating mode of the PiezoSmart sensor allowing the program editor software to read or add information contained in the memory chip.

Kistler publication K20.302 "IEEE P 1451.4 Measurement With Smart Transducers includes information on the IEEE Standard 1451.4, the operating manner of smart sensors, a detail review of the TEDS Editor and data sheet content. Call Kistler to request a copy.

3.1 Supplied Items

All accelerometers designed for screw mounting or stud mounting are supplied with appropriate mounting screws or mounting studs. Units intended for adhesive mounting are provided with Petrowax. The supplied accessories are included in the unit specifications.

4 Technical Information, Functional Description

4.1 Piezoelectric Measurement Concept

Piezoelectric accelerometers convert acceleration into an electric charge. The charge derived by the accelerometer is proportional to the force acting on the internal quartz (piezoelectric) element. Accordingly, the mechanical variable (acceleration) is derived from a force measurement.

4.2 K-Shear Background

The sensing assembly (See Figure 1) consists of a center post, quartz piezoelectric crystals, seismic mass and a preload bolt. Since the unit is operated in a shear arrangement it will sense motion perpendicular to the base. When the accelerometer is attached to a vibrating structure, the mass exerts a shear force on the quartz piezoelectric crystal. This applied force causes the piezoelectric material to produce an electric charge. Since force is mass times acceleration (from Newton's second law), the charge produced is proportional to acceleration, since m is constant. This is represented by:

$$a = \frac{F}{m}$$

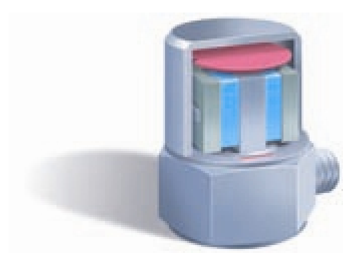

Figure 1: Inside View of K-Shear Single Axis Accelerometer (Shown for Simplicity).

The K-Shear sensing element offers many advantages over previous shear and compression mode designs. Because of shear construction the accelerometer is insensitive to thermal transients, transverse (cross-axis) motion and the effects of base strain. This insensitivity to activities happening in an adjacent axis minimize cross talk effects in these triaxial accelerometers. In a triaxial accelerometer three sensing elements, one per axis is used.

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4.3 Low Impedance Output

Contained within the accelerometer housing are three miniature electronic circuits. These circuits convert the high impedance charge signal generated by each of the piezoelectric elements into a low impedance voltage output signals each with output impedance typically below 100 Ohms.

The integral impedance converters are powered by an external power source (coupler) that uses a four-wire cable between the accelerometer and coupler. The signal and power share a common line for each axis. The coupler provides a constant current source to the accelerometer and decouples the DC bias (see Section 6.1.3) from the measuring instrument. The useful signal is seen as a varying voltage over an 11 VDC (nominal) bias.

Low impedance accelerometers are ideally suited for applications where long or moving cables are required or in high humidity or other contaminated atmospheres. They eliminate the problems associated with a high impedance output by providing a voltage signal with low impedance and a high frequency response. The calibration factors for these accelerometers are given in mV/g.

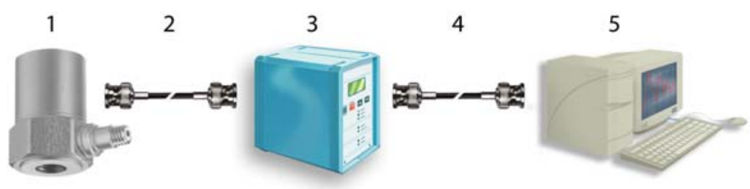


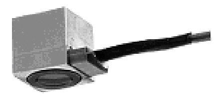
Figure 2: Diagram of a Triaxial Acceleration System

1. Triaxial accelerometer
2. Breakout cable (extension cable also avaliable)
3. Multichannel Coupler
4. Cable
5. Read out equipment

5 Installation

5.1 General

For proper operation of K-Shear triaxial accelerometers, care must be taken during the installation process. Careful installation will result in optimal high frequency response, accuracy and reliability. Dimensional characteristics for each accelerometer can be found on their respective data sheets.



8790 adhesive mounted miniature cube , with integral cable

8791 adhesive mounted miniature cube , with integral cable

8792 screw mounted rugged, low profile accel

8793 screw mounted, for low and high temp applications

8794 screw mounted low profile accel with integral cable

8795 stud mounted cube, for low and high temp applications

5.2 Surface Preparation

A smooth, flat surface is necessary for both screw and adhesively mounted accelerometers. If the surface is not completely flat, the coupling between the accelerometer and the test article introduces distortion into the measurement. A rough surface creates voids between the mounting surfaces that reduce high frequency transmissibility.

For optimum frequency response, the surface and hole preparation should be according to the instructions in Table 4. The roughness should not exceed 32 microinches (0.8 micrometers).

5.3 Screw Mounting

K-Shear triaxial series 8792A…, 8793A… and 8794A… are specifically designed for screw mounting. Most Kistler mounting screws are machined from Beryllium Copper for high strength and low modulus of elasticity, coupled with high elastic limits. The following guidelines should be followed when screw mounting accelerometers.

1. Drill and tap an adequate hole to ensure a flush mount of the accelerometer. Make sure the screw does not bottom out and firmly secures the accelerometer. A chamfer should be machined at the top of the mounting hole to ensure that the base of the accelerometer makes full contact with the mounting surface (See Table 3).
2. Completely clean the surface prior to mounting.
3. Apply a thin coat of silicon grease to both the accelerometer and mounting surface.
4. Always use the proper sockets and a torque wrench when installing accelerometers. Tighten the accelerometer to the torque specified on the individually supplied calibration certificate or as specified in Table 2. Do Not Over Tighten.

in-lbs	Nm	Accelerometer Types
18 ±2	2 ±0.2	8792
4 ±1	0.45 ±0.1	8793, 8794

Table 2: Recommended Screw Mounting Torque

5.4 Stud Mounting

K-Shear triaxial accelerometer series 8795A… is provided with a 10-32 stud hole for use with a stud. The type 8402 mounting stud is included for use when stud mounting is desired. This stud is removable once installed allowing the unit to be adhesively mounted.

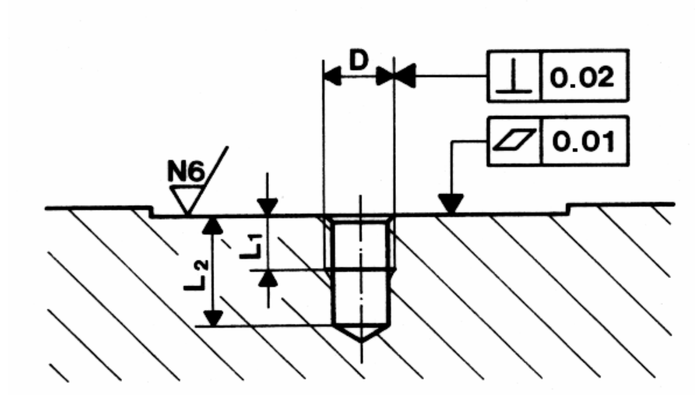
The furnished stud is machined from Beryllium Copper for high strength and low modulus of elasticity, coupled with high elastic limits. Surface preparation considerations from Section 5.2 should be followed. The following guidelines should be followed when stud mounting the series 8795A….

1. Drill and tap an adequate hole to ensure a flush mount of the accelerometer. Make sure the stud does not bottom out and firmly secures the accelerometer. A chamfer should be machined at the top of the mounting hole to ensure that the base of the accelerometer makes full contact with the mounting surface (See Table 4).
2. Completely clean the surface prior to mounting.
3. Apply a thin coat of silicon grease to both the accelerometer and mounting surface.

4. Always use the proper sockets and a torque wrench when installing accelerometers. Tighten the accelerometer to the torque specified on the individually supplied calibration certificate or as specified in Table 3. Do Not Over Tighten.

in-lbs	Nm	Accelerometer Type
18 ±2	2 ±0.2	8795

Table 3: Recommended Stud Mounting Torque



Stud Type	D	L1 mm	L2 mm
8402	10-32 UNF	4.0	8.0
8411	M6	8.0	14.0

Table 4: Stud Mounting Preparation

5.5 Direct Adhesive Mounting

Some K-Shear triaxial accelerometers are specifically designed for adhesive mounting such as the 8790A500 and 8791A250M01 and require no special mounting adapters. Units furnished with stud holes, like the series 8795A…, can also be used with adhesives.

The surface should be smooth and flat. A cyanoacrylate type adhesive such as Eastman 910, Loctite 496, super glue is recommended. While epoxies can also be used, cyanoacrylate adhesives provide an extremely thin bond providing optimal frequency response.

Adhesive preparations may not be suitable for applications at elevated temperatures. Upper temperature operating limits should be observed as adhesive failure could result in the unintentional release of an accelerometer that could subject the instrument to damage.

When adhesive mounting an accelerometer with a tapped hole, make sure that no adhesive is allowed to enter the hole. This could make stud mounting difficult or impossible at some future time.

Remove the sensor with a manufacturer's recommended adhesive solvent. Acetone is effective for the removal of cyanoacrylate adhesives. Use the proper size wrench to twist the sensor loose and break-off the adhesive. Do Not Impact!

Warning! Do not attempt to remove triaxial accelerometers by twisting with pliers, a wrench or impacting. Applying torque with a wrench or other tools will damage the aluminum housings. Remove these accelerometers using a recommended cyanoacrylate solvent (e.g., Loctite 768), then twist with fingers. If conditions permit, Petro-Wax (see Section 5.6) is the ideal mounting material for these sensors.

5.6 Mounting with Wax

Bee's wax has been used as a mounting agent for many years. The provided Petro-Wax (Kistler Type 8432 or P/N P102 from Katt and Associates; P.O. Box 623, Zoar, Ohio 44697 or equivalent) is a good replacement for bee's wax since it has been formulated to provide improved frequency response. Wax is a good mounting agent for lightweight sensors in temporary installations where near room temperatures are encountered.

5.7 Magnetic Mounting

Magnetic mounting is a convenient way to take measurements on ferro-magnetic surfaces. Magnetic mounting adds considerable mass and reduces high frequency response.

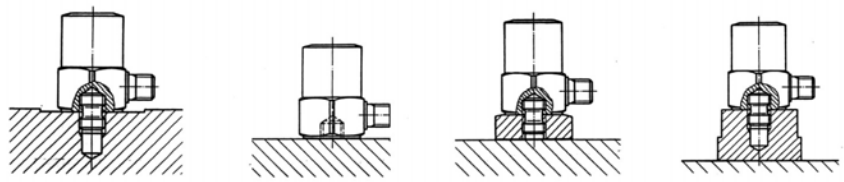
Make sure that the mounting surface is flat and clean. An oil or grease film greatly enhances the coupling, hence improving the working frequency range.

Table 5 shows the magnetic base available from Kistler along with its characteristics. In order to achieve the specified holding strength a thin grease film should be applied to the mounting surface.

Sensor Type	Magnet Type	Holding Strength	Holding Strength	Upper Frequency Limit (± 5%)	Mass
		N	lbs	kHz	g
8795A	8452A	53	12	4	18

Table 5: Magnetic Mounting Specifications

5.8 Summary of Mounting Methods



Stud Mount Adhesive Mount Magnetic Mount

Mounting Method	Type	Advantages	Disadvantages	Remarks
Stud	See Section 8.1	best coupling of sensor to test specimen for highest frequency response	requires threaded hole in specimen	control mounting torque, use Silicone grease
Adhesive Mounting Pad	See Section 8.2	allows stud mounting, provides electrical isolation	lowers resonant frequency	pads are usually epoxied to test specimen
Adhesive Wax	8432 Petro Wax, bees wax	ideal for light weight units	limited temperature range, not suitable for larger sensors	
Adhesive Cement	Loctite, Permabond, (Cyanoacrylate)	good, strong coupling of sensor to specimen, higher temperature capability than wax	difficult to remove sensor, requires solvents	suited for more permanent applications and high frequency measurements
Adhesive Tape	"Double Stick", etc.	easy application	only for low g, low weight accelerometers	only for very limited applications
Magnetic Base	See Table 4	easy and quick instal-lation	adds considerably to mass loading, lowers resonant frequency	limited to ferro-magnetic materials

5.9 Securing Cables

Figures 3 and 4 show the correct and incorrect ways for installing cables. Allow a sufficient radius to ensure a proper strain relief. The actual radius will depend on the cables being used. It is recommended that cables be secured to the vibrating surface to minimize cable and connector fatigue failures. Secure cables with a cable clamp. Tape is acceptable on temporary installations.

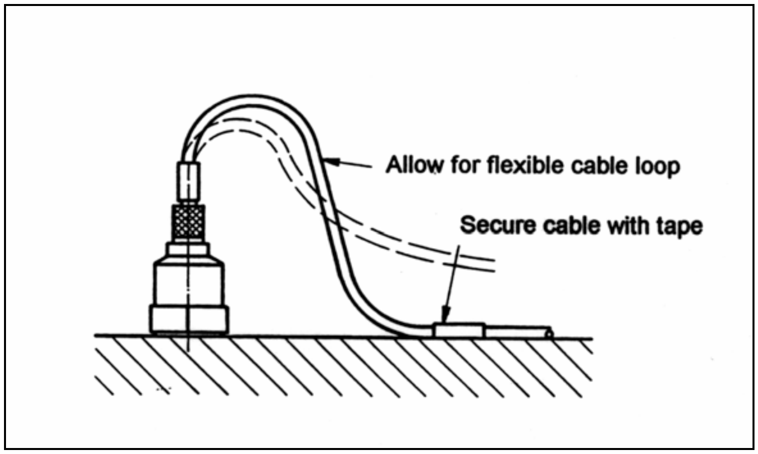


Figure 3: Correct Cable Strain Relief

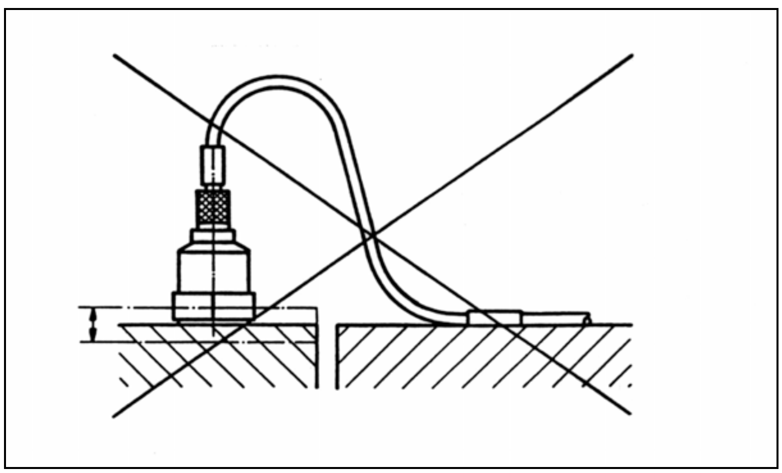


Figure 4: Incorrect Cable Strain Relief

See Section 8.5 for information on cables available from Kistler.

5.10 Mass Loading Effects

When a device such an accelerometer is attached to a test article the acceleration to be measured will be altered. These effects can be reduced to an insignificant amount by the proper selection of the measuring accelerometer. It can generally be assumed that the presence of an accelerometer does not significantly affect the system response provided the dynamic mass of the accelerometer is considerably less than the test specimen. As a general "rule-of-thumb", the weight of the accelerometer should be no greater than 10% the weight of the test specimen. An example of possible accelerometer induced distortion is the case of a heavy sensor mounted in the center of a thin printed circuit board.

6 Operation

Piezoelectric accelerometers are self-generating transducers. While the quartz piezoelectric sensor does not require an external power source, it is necessary to provide power to the K-Shear's internal electronic impedance converter. This section is intended to provide the user with the information necessary to ensure accurate measurements with K-Shear accelerometers. Topics to be covered include powering, signal conditioning, frequency response and driving long cables.

6.1 Powering

6.1.1 Using "Built In" Power Sources

Many FFT analyzers and vibration monitors are available with internal accelerometer power supplies. Often identified as "IEPE" or "Constant Current Sources", these internal power supplies are generally compatible with Kistler K-Shear triaxial accelerometers.

Before using the built-in power source, compare the measurement instrument current source specifications with the current and voltage specifications of the K-Shear triaxial accelerometer to be used. If the instrument accelerometer power is within the range required by the specific K-Shear triaxial accelerometer there should be no problem with compatibility. If the user plans to drive long cables (over 430 feet/130 m) the guidelines in section 6.2 should be followed. The accelerometer low frequency response may be affected by the input impedance of the measuring instrument as discussed in Section 6.4.4.

Caution: Many industrial monitors have adjustable current controls. Exceeding the maximum current rating of any accelerometer may cause permanent damage and void the warranty.

6.1.2 Kistler Couplers

Kistler offers a comprehensive line of power supply/couplers and dual mode amplifiers. If the user is utilizing these instruments, they are designed for compatibility with K-Shear triaxial accelerometers. As each accelerometer axis is isolated from the other two, each axis is an individual channel. Thus a minimum of three single channel power supply/couplers or a multichannel coupler is required for a single triaxial accelerometer. The chart below provides a listing of available power supply/couplers. Please request data sheets on specific models.

Type	Description
5108A	Passive coupler. Customer provides power. CE approved
5110	Battery powered single channel coupler with banana plugs for direct connection to a multimeter and BNC connection. CE approved
5114	AC/battery powered single channel coupler with LCD display to monitor bias. LED's indicate status of sensor circuit. Portable or tabletop use. CE approved
5118B2	AC/battery powered single channel coupler with 1X, 10X, 100X gain and high pass filter switch. CE approved
5134A	4-channel AC powered coupler and amplifier. Selectable high pass filters and gain. Random noise generator for testing with FFT analyzer. Remote control and sensing via RS-232C. CE approved
5148	16-channel rack mountable AC powered coupler with unity gain. LED's indicate circuit integrity. Front panel BNC output connectors.

6.1.3 The Constant Current Power Supply/Coupler

To better understand the purpose of the current source, a review of the Piezotron impedance converter is in order. For simplicity, a single axis is described.

Figure 5 is a simplified schematic diagram of the K-Shear accelerometer. When excited, the quartz element produces a charge proportional to the measurand. The resultant voltage (developed across Cr in Figure 5) is applied to the input of the internal impedance converter (Electrometer Amplifier). The impedance converter will produce an output voltage that follows the input faithfully.

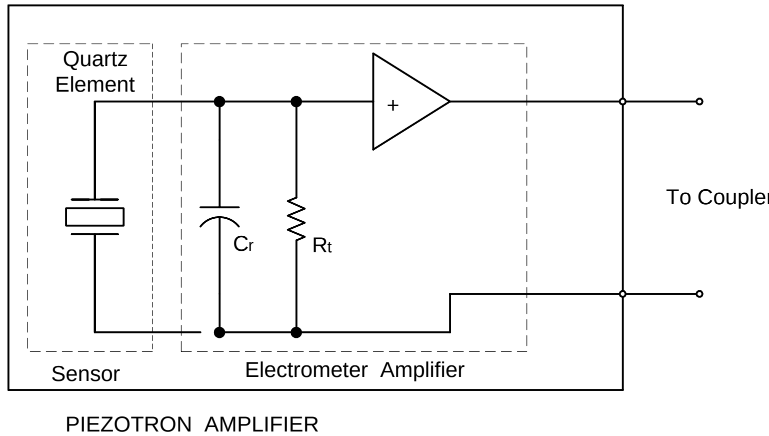


Figure 5: Simplified Diagram of One Axis of a K-Shear Triaxial Accelerometer

The source resistor (Rs), while located in the coupler, is a part of the source follower circuit. In order for the Piezotron to provide enough gain, Rs must have a high dynamic resistance while being able to provide enough current (low DC resistance) as not to affect the high frequency response of the system. Replacing Rs with a constant current diode (Figure 6) provides a device with a high dynamic resistance while allowing sufficient current to flow.

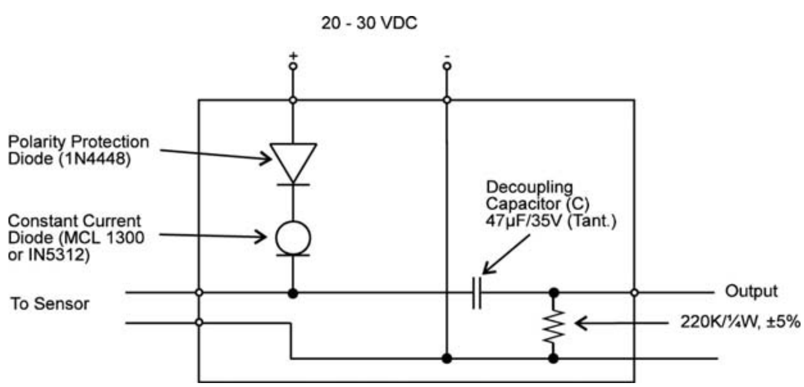


Figure 6: Schematic Diagram of A Simple Coupler

Figure 6 shows a schematic of a simple, single channel power supply coupler for K-Shear accelerometers. DC power is supplied from a 24-volt source such as a regulated power supply or batteries. One advantage of the Piezotron system is the fact that a simple two-wire coaxial cable is used for both power and signal. Because the signal and power both share the same line, it is necessary to include the capacitor C to decouple (or block) the DC from the measurement instrument input.

Operating within the voltage range of 20...30 VDC assures a full undistorted ±5 Volt output amplitude (See Figure 7). If the source voltage is reduced, distortion and clipping of the signal will occur if one attempts to use the full amplitude range of the accelerometer (See Figure 8). The following equations are provided to calculate the maximum full-scale voltage available when operating with a reduced voltage.

Maximum positive amplitude:

$$+Fs = (Es - 1) - Eb$$

Maximum negative amplitude:

$$-Fs = Eb - 2$$

Where:

F*s* = Full Scale output voltage (peak).

E*s* = DC supply voltage (sometimes called compliance voltage).

E*b* = bias voltage (from calibration certificate).

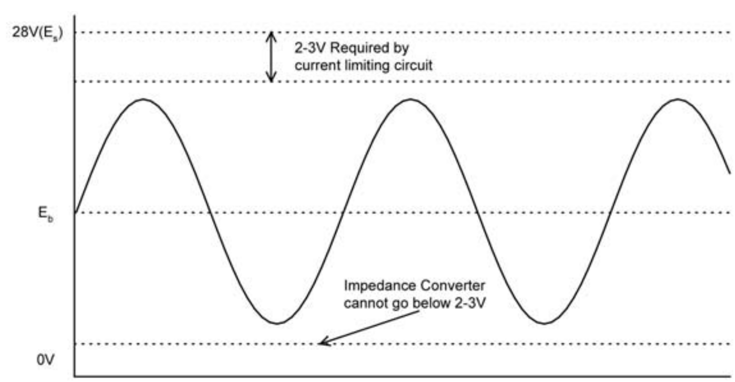


Figure 7: Output Signal When Operated With 28V Supply

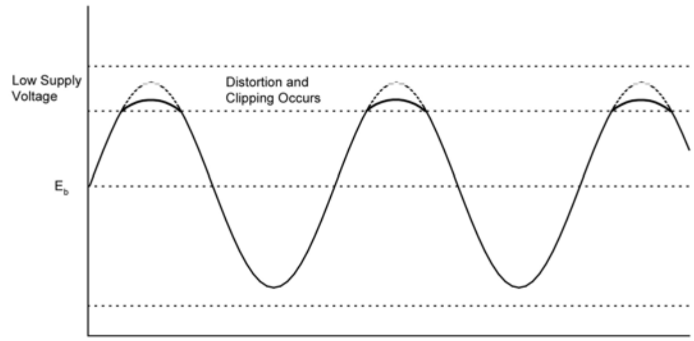


Figure 8: Example of Output Distorting and Clipping when Supply Voltage is Low

6.1.4 Sensor Power-Up

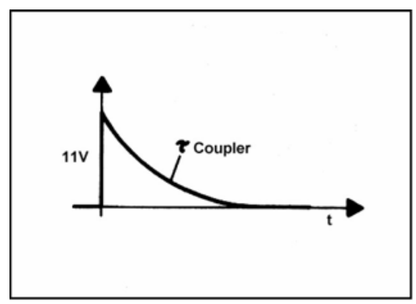


Figure 9: Behavior of The Output Signal During Power-Up

When power is first applied to the sensor, (through an AC coupled coupler) a voltage will appear at the output of the coupler (See Figure 9). As the coupling capacitor discharges, the DC output, from the coupler will drop to zero Volts. This "settling time" is equal to 5 times the time constant of the coupler employed. Allow time for the unit to "settle" before making measurements.

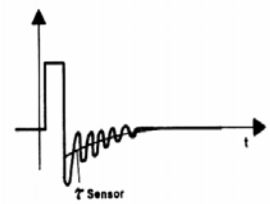


Figure 10: Output Behavior During Overload and Recovery

6.1.5 Overload Recovery

Whenever the impedance converter is driven by a signal exceeding the normal operating range, certain components will become non operational. During this non-operational state, the amplifier components are protected from overload damage. The amount of time required for recovery from an overload depends on several factors. Important for overload recovery time is the size of the overload. As with power-up, the time constant makes the biggest contribution to the recovery time. Figure 10 illustrates a typical overload and recovery sequence.

6.1.6 Supply Voltage Effects

Long-term fluctuations in the power supply level between 24 and 30 volts can be tolerated. The sensitivity shift caused by such deviation is less than 0.05%/Volt. A normal noise level is maintained if the power supply ripple is 25mV RMS or less. The polarity must be maintained throughout the system since applying reversed polarity power may cause damage to the accelerometer.

6.2 Driving Long Cables

The voltage mode Piezotron circuit allows for long cable runs with low noise susceptibility. Most laboratory instruments with built-in accelerometer power provide current in the range of 2 to 4 mA. Most Kistler couplers are set at the factory to provide 4mA of source current. 4mA is a good compromise value for maximum frequency response and high reliability.

Assuming a cable capacitance of 30pF/foot(98 pF/m), the full frequency range of K-Shear accelerometers can be realized (±5 Volt output) up to a length of 430 feet (130 meters) with 4 mA of drive current. Most Kistler cable types and common RG58 coaxial cable have a rated capacitance of 30 pF/foot (98 pF/m). For most laboratory applications this is quite adequate.

As cable length is increased the cable capacitance becomes significant, thereby loading the Piezotron impedance converter. If the current is not sufficient to charge the cable at an adequate rate, high frequency distortion will be experienced. The solution to this problem is to increase the drive current

For the user's convenience a chart is provided (See Table 6). The values given are based on a cable capacitance of 30 pF/foot (98 pF/m). The list of cables available from Kistler can be found in section 9.3. All cables in this section are 30pF/ft (98 pF/m). except for the 1331, 1635 and 1639 which are 20 pF per foot(65 pF/m).

Maximum Frequency **+/- 5%**	**Cable Length** **Ft (meters)**	**Current (mA) Required** **For Output Signal +/- 1 Volt**	**Current (mA) Required** **For Output Signal +/- 5 Volt**
10 kHz	1000 -300 2000 -600	2 4	10 18
9 kHz	1000 -300 2000 -600	2 4	9 17
8 kHz	2000 -600 2000 -600	2 3	8 15
7 kHz	1000 -300 2000 -600	2 3	7 14
6 kHz	1000 -300 2000 -600	2 3	6 12
5 kHz	1000 -300 2000 -600	2 2	5 10

*Based on 30 pF/ foot (98 pF/meter)

Table 6: Current Requirements for Driving Long Cables

The current requirements can be calculated using the following equation:

$$I = 2\pi fCE$$

Where:

I = Current in Amperes
f = Frequency in Hz
C = Capacitance in Farads
E = Output in Volts, Peak

Never use more current than required to make the desired measurement. Never exceed the maximum current.

If a requirement calls for 1,000 feet (300m) of cable and the maximum frequency of interest is only 5 kHz than the user should select (From Table 6) 5 mA of current rather than the 10 mA required for the full 10 kHz range. As current is increased, the noise produced by the Piezotron circuit will also increase. The miniature Piezotron amplifier will produce more heat as current is increased resulting in reduced life and reliability.

The specified maximum temperature range (in the data sheet) is based on 4 mA of supply current. It is necessary to degrade the maximum operating temperature when operating under increased temperature conditions. The graph in Figure 11 shows the approximate maximum temperature vs. operating current.

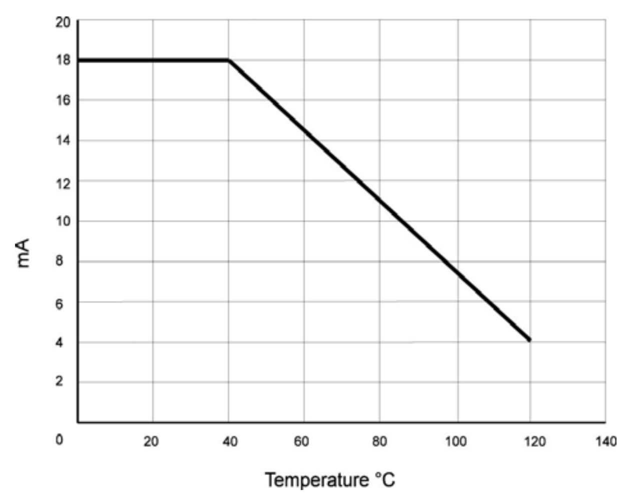


Figure 11: Maximum PIEZOTRON Current as a Function of Operating Temperature

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The equation for computing the maximum current for a specific temperature is:

$$I_{\max} = 0.16(150 - t_{\max})$$

where:

I = Current in mA

t = Temperature is C

6.3 Ground Loops

The ground loop is one of the most common causes of ground noise. It will usually manifest itself as unwanted noise occurring at the line frequency (50 - 60 Hz).

Ground loops occur when current flows between the accelerometer and coupler/conditioner grounds. If the coupler ground and the accelerometer ground are at exactly the same potential, there will be no ground loop between these two devices. If the ground potential is different between the accelerometer mounting surface and the coupler ground, a slight current will flow causing a ground loop to occur (See Figure 12). While the amount of current might seem insignificant, the measurement levels are generally very low and the ground current will appear on the signal at a measurable level. There can be many causes of ground loops in a complete measurement system but the ground path between the accelerometer and coupler is among the most common.

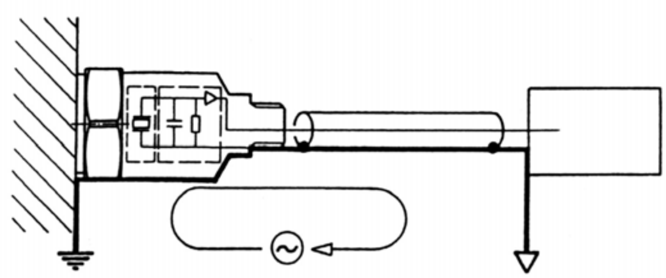

Figure 12: Current Path (Loop) Between the Coupler and Accelerometer Ground

One method used to eliminate ground loops is to isolate the ground of the accelerometer from the ground of the test article (See Figures 13 and 14). K-Shear accelerometer models 8790A…, 8792A… and models with the suffix "M1" or "M3" are ground isolated; therefore, there should be no ground currents flowing between the accelerometer or other devices.

Kistler offers isolation mounting studs for some non-isolated accelerometers. See section 8.1 for details.

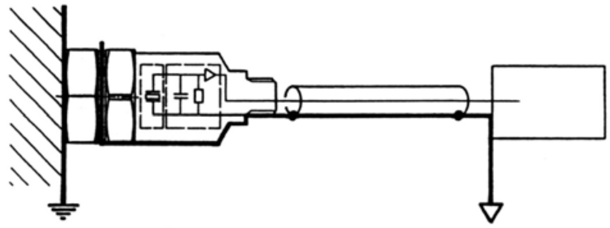

Figure 13: The Current Path is Broken with the use of a Ground Isolation Pad

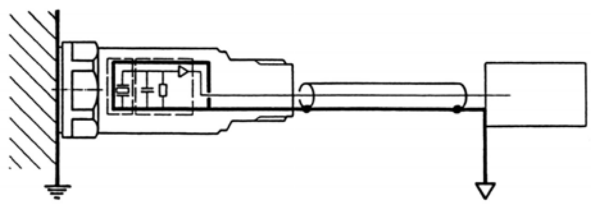

Figure 14: Isolated Accelerometer Eliminates Current Path

6.4 Frequency Response Limits

All accelerometers have specified frequency limits, which are a function of the mechanical and electrical design. In order to better understand the measurement process and limitations this section will describe both the inherent limitations and limitations imposed by the installation, operation and measurement-analysis instrumentation.

6.4.1 Definition and Frequency Response Standards

Frequency response is defined as a maximum specified amplitude variation, from an established reference frequency, over a specified bandwidth. K-Shear accelerometers are specified with maximum variations of ±5% to ±10%. Kistler uses the industry standard reference frequency of 100Hz.

Many users may be more familiar with the common frequency response standard for electronic equipment, which is generally 3dB. In most mechanical applications an error of 3db (about 30%) does not provide the required accuracy; thus, 5% (10% in some cases) is considered a more useful and accurate limit.

6.4.2 High Frequency Limitations

The inherent high frequency limit of an accelerometer is a function of its mechanical characteristics. K-Shear accelerometers can be represented as an undamped single degree-of-freedom spring-mass system. Accelerometers are modeled by the classical second order differential equation whose solution is:

$$\frac{a_o}{a_b} \cong \frac{1}{\sqrt{\left[1-\left(\frac{f}{f_n}\right)^2\right]^2+\left(\frac{1}{Q^2}\right)\left(\frac{f}{f_n}\right)^2}}$$

Where:

Where:

$$\begin{aligned}
&f_n = \text{undamped natural (resonant) frequency (Hz)}\\
&f \ = \text{frequency at any given point of the curve (Hz)}\\
&a_o = \text{output acceleration}\\
&a_b = \text{mounting base of reference acceleration} \left(\frac{f}{fn}=1\right)\\
&Q = \text{factor of amplitude increase at resonance}
\end{aligned}$$

Quartz accelerometers have a Q of approximately 10 to 40 and therefore the phase angle can be written as:

$$\text{Phase Lag (deg)} \cong \frac{60}{Q}\left(\frac{f}{f_n}\right) for \frac{f}{f_n} \le \frac{2}{5}$$

$$\text{of a single degree of freedom system}$$

Using the basic equation for resonance one can better understand the basic accelerometer mechanical system and the affect on high frequency response.

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

The spring constant (k) is determined by the stiffness of the quartz crystal and preload stud. The mass (m) is the supported part of the accelerometer structure. This spring-mass system is the basic component that converts the input forces into acceleration. As with any single degree-of-freedom system, a resonance condition will occur (per the above equation) which causes the sensitivity to increase with frequency.

It is desirable to have a uniform frequency response over the specified frequency range. As can be seen, (Figure 15), the sensitivity greatly increases when approaching resonance. About a 5% amplitude rise can be expected at 9/40 of the resonant frequency. It is possible to operate above the maximum specified frequency by applying the appropriate correction factors, or filtering to compensate for the resonant rise in response.

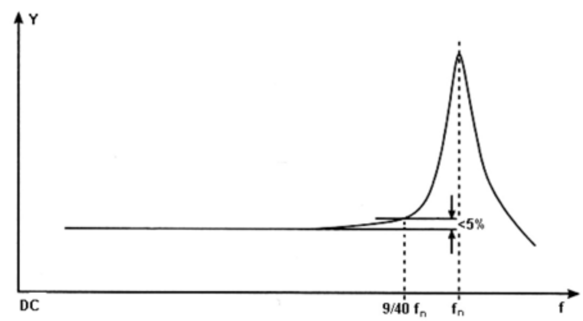

Figure 15: Typical Frequency Response of a Piezoelectric Accelerometer

Section 5 provides important information regarding mounting. It is important that these instructions are followed to ensure optimal transmissibility at high frequencies. Mounting to rough surfaces, using soft adhesives and magnetic mounts will limit the high frequency range to some value lower than the specified frequency range.

6.4.3 Low Frequency Limitations - Transducer

The low frequency response of accelerometers is an electrical limitation. The K-Shear design is such that low frequency drift caused by temperature variations is virtually eliminated.

The K-Shear sensing element is a self-generating device, which produces a voltage proportional to the applied acceleration. The piezoelectric output voltage is applied to the high impedance input of the FET where it is converted to a low impedance output voltage signal.

At low frequencies the amplifier input circuit acts as a single order high-pass filter (See Figure 5) whose amplitude and phase are:

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$$\frac{V_o}{V_{in}} = \frac{2\pi f(\tau)}{\sqrt{1+[2\pi f(\tau)]^2}}$$

$$\text{phase lead (deg)} = \text{arc tan } \frac{1}{2\pi f(\tau)} \cong 80\sqrt{\frac{V_{in} - V_o}{V_{in}}}$$

The high pass filter has a time constant (t) as determined by the capacitance and resistance of the internal amplifier gate circuit. The time constant is defined as the time required for the capacitors to reach 63.3% of full charge as determined by t=RC. As capacitance and/or resistance is increased, the time constant becomes longer thus an increase in the low frequency response since frequency is the reciprocal of time.

The low frequency limit of Kistler K-Shear accelerometers is defined as the point where the response is -5% (also -10% where specified) below the reference point (100 Hz). The low frequency specification is a nominal value. The actual low frequency point of your specific accelerometer can be determined from the measured t found on the unit's calibration certificate. The following equation can be used to determine where the low frequency response will be 5% below reference

$$f_{-5\%} = \frac{0.5}{\tau}$$

where:

f = frequency in Hz

t = time in seconds (from calibration certificate)

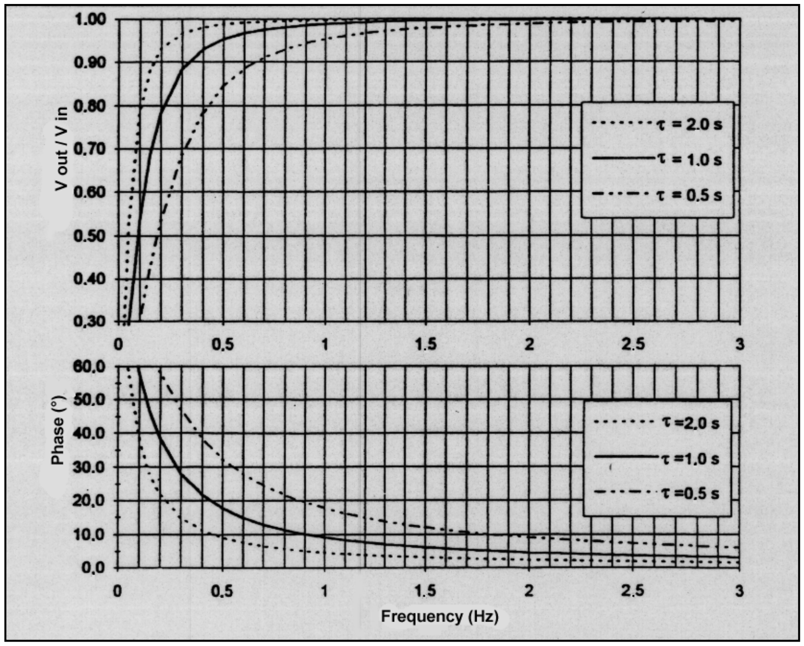


Figure 16: Frequency and Phase Response of K-Shear Accelerometers at Low Frequencies are Dependent on the Time Constant

6.4.4 Low Frequency Limitations - Coupler and Readout Instrumentation

When making low frequency measurements with K-Shear accelerometers it is important that the user consider the affects of the coupler and measurement instrumentation on the low frequency response.

If the coupler selected includes an isolation amplifier, the low frequency cutoff will be that specified in the coupler data sheet and the readout instrument's specifications. The actual low frequency limit will be the higher of the two. As an example, an FFT analyzer is being used that has a low frequency cutoff of 0.5Hz and a coupler with a 0.07 Hz low frequency cutoff is being utilized. The low frequency response limit will be 0.5 Hz.

If DC methods of bias decoupling are utilized, the transducer's time constant will be unchanged thereby providing optimal low frequency response.

When using a simple coupler circuit (Figure 7), with AC coupling, a time constant equal to the product of the decoupling capacitor's (C) capacitance times the readout instrument's input impedance will be produced. Using the formula t=RC (The readout instruments impedance = R) it can be seen that the lower the readout instrument's input impedance (Z), the shorter the time constant thus degrading the low frequency response. In systems with AC bias decoupling the time constant can be approximated by taking the product of the transducer and coupler time constants and dividing by their sum.

When using a simple coupler (Figure 7) the user should be aware that the system formed by the coupler and readout instrumentation can shorten the time constant thus degrading the low frequency response. The new system time constant can be approximated by the equation:

$$\tau = \frac{R_1 \text{ x } R_2}{R_1 + R_2} C$$

Where:

R_1 = Output resistance of the coupler

R_2 = Input resistance of the readout instrument

C = Capacitance of the Coupler's output capacitor

7 Maintenance and Calibration

7.1 General

The sealed construction of K-Shear accelerometers precludes any internal maintenance or repairs. External cleaning is important as covered in section 7.4.

7.2 Trouble Shooting

Should the user experience any problems with a K-Shear accelerometer, the following procedure should be followed.

Check for possible intermittent cables. Faulty cables are the most common cause of accelerometer measurement problems. The application of a test signal at the accelerometer end of the cable is an effective means to determine if the problem is accelerometer, cable, or measurement instrument related.

Check for possible radio-frequency radiation in certain models. Types, 8793A… and 8794A…, if subjected to operating conditions inducing radio-frequency current in the cabling, may be affected by higher than normal noise levels in the output signal. Several remedial actions, such as the addition of ferrite chokes to the cable, may remedy this condition.

Kistler couplers are equipped with a bias monitor, which can be used to check the cable integrity and detect malfunctions in the accelerometer. Use the bias monitor as follows:

A. With the accelerometer connected, check for a normal reading on the bias indicator (normal should be around 11 VDC).

B. If the bias reading is high:

  1. Disconnect the accelerometer at the accelerometer end of the cable.

  2. If the bias indicator reading decreases; the accelerometer is defective.

C. If the bias indicator reads low, disconnect the accelerometer

  1. If the bias indicator still reads low, replace the cable.

  2. If the bias level changes when the cable is disconnected, the accelerometer is defective.

Analysis and measurement equipment with built-in accelerometer sources generally do not have bias monitors. In these cases, the user can attach a T connector to the measurement/analysis instrument and measure the actual bias voltage. If the system is operating properly, the bias voltage should measure approximately 11 Volts. Defects should show-up as in A and B above.

If it is determined that the accelerometer requires repair, please contact the factory or your local distributor/representative for repair information.

7.3 Repairs

Field repair is not possible. Please return the accelerometer to the factory together with a statement of problems encountered and unit serial number. Recalibration is recommended when returning an accelerometer to the factory. Please contact the factory and obtain a return authorization number prior to return.

7.4 Cleaning

Clean the transducer with Isopropyl alcohol and lint-free paper wipes. Never use air-blast to clean the connector; it may deposit a water vapor film. Excess cyanoacrylate adhesive (e.g. Eastman 910, Loctite 430, etc.) on the mounting surface may be removed with dimethylformamide or acetone. Refer to the adhesive manufacturer's product data for recommended removal agent.

Do not us an abrasive on the base surface. This can affect the flatness thereby reducing high frequency transmissibility.

7.5 Calibration

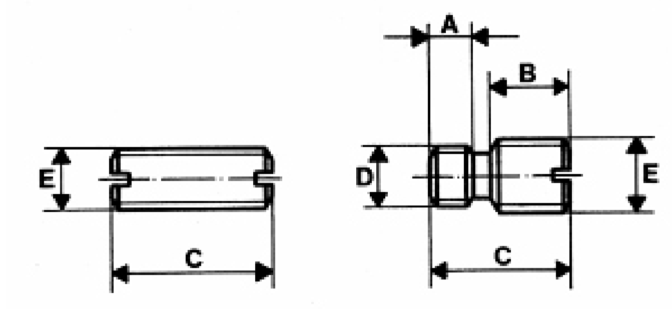
Calibration consists of the precise determination of the accelerometer sensitivity by direct comparison, at various frequencies, to a National Institute of Standards and Technology (NIST) traceable standard accelerometer. The test accelerometer is attached to the top of the standard (Kistler type 8676K) in a "back-to-back" configuration. The standard and test accelerometer are excited at appropriate frequencies and levels. The resultant amplitudes of both accelerometers are then compared to determine if the test unit is within tolerance.

Calibration services are available from Kistler. Kistler acceptance test instrumentation is in conformance with MIL-STD-45662A. A calibration certificate will be supplied showing calibration results from standards traceable to NIST.

8 Accessories

8.1 Studs

The following studs can be used with Kistler accelerometers:



Type	A	B	C	D	E	Used With
8402	0.10 [2.5]	0.10 [2.5]	0.28 [7.1]	10-32 UNF	10-32 UNF	8795

(Dimensions in mm)

8.2 Magnetic Mounting Bases

See Table 3 For Magnetic Mounting Specifications.

8.3 Cables

All K-Shear accelerometers with detachable cables are furnished without cables. Types 8790A… and 8791A… are furnished with integral cables. For interface or extension length, an additional cable is required. The suffix sp(X) indicates that custom lengths are available. Certain standard lengths are stocked. Call Kistler for special cable requirements.

Cable Type	Cable Description
1761sp(x)	General purpose cable/connector. 10-32 pos. to BNC pos. Accelerometer output cable.
1756B(X)	Interface cable/connector. 4-pin Microtech neg. to BNC pos. Triaxial accelerometer cable.
1578Asp(X)	Extension cable/connector. 4-pin Microtech neg. to 4-pin Microtech pos. Triaxial accelerometer cable.

Table 7: Cables used with K-Shear triaxial accelerometers

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Accelerometer Type	Associated Cable Type
8790 (has integral output cable)	suggested 3 each 1761sp(x)
8791 (has integral output cable)	requires interface cable 1756sp(X)
8792, 8793, 8794, 8795	1756B(X) and 1578A(X)

Table 8: K-Shear accelerometers with associated cable

9 Warranty

Regarding the warranty reference is made to the agreement between the respective contracting parties.

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